

SENIOR SOFTWARE / AI ENGINEER

ROMAN GIRIN, Ph.D.

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PROFESSIONAL SUMMARY

Senior Software & Applied AI Engineer with 20+ years of experience specializing in high-load backend systems, enterprise cloud migrations, and production-grade AI infrastructure. Combines an advanced academic foundation (**Ph.D. in Applied Neural Networks & Instrumentation**) with decades of hands-on mastery across the .NET ecosystem, microservices, and applied machine learning (Python, PyTorch, LLMs). Focused entirely on software engineering and AI integration rather than theoretical data analysis or statistical research.

KEY SKILLS & TECH STACK

- **Programming Languages:** C#, Python, SQL (T-SQL, PL/pgSQL), C++
- **Platforms & Frameworks:** .NET 8 / Core / Framework, ASP.NET, Entity Framework Core, Django REST Framework
- **Software Architecture & Patterns:** Monolith-to-Microservices, Domain-Driven Design (DDD), Event-Driven Architecture (EDA), CQRS, Saga, Outbox, RESTful API, HATEOAS, SOLID, GoF
- **AI Orchestration & Systems Integration (.NET):** MS Semantic Kernel, RAG, Vector Search (`pgvector`), Semantic Routing, LLM Agents
- **Applied ML & MLOps Frameworks (Python):** PyTorch, Hugging Face Transformers, LoRA Fine-Tuning, OpenCV, TorchVision, Scikit-Learn
- **Message Brokers & Streaming:** Azure Event Grid, Azure Queues, Kafka, RabbitMQ
- **Databases:** MS SQL, PostgreSQL, noSQL Azure Table Storage
- **Cloud, DevOps & Observability:** Docker, Azure Kubernetes Service (AKS), Azure Container Apps, Azure DevOps (CI/CD pipelines)
- **Testing & Engineering Culture:** xUnit, NSubstitute, Test-Driven Development (TDD)
- **AI Productive Tools:** Claude Code, LM Studio

PROFESSIONAL EXPERIENCE

River Logic, Inc. | Seattle, USA (Remote)

Senior Modeling Software .NET Developer / AI Engineer | **March 2014 – Present**

- **Architecture Modernization:** Spearheaded the architectural transition from a legacy 2-tier to 3-tier architecture. Migrated Business Logic Layer (BLL) from database-tier (MSSQL T-SQL) into the .NET application layer (C#), improving system maintainability resulting in a 30% reduction in support ticket volume. Implemented compatibility with PostgreSQL.

- **Windows to Linux Migration:** Spearheaded the core codebase migration from .NET Framework to cross-platform .NET 8, implementing a modern asynchronous programming model (`async/await`) that slashed system resource utilization by 15%.
- **Distributed microservices in Cloud:** Decomposed an on-premise monolith into high-load microservices deployed on Azure Container Apps and Azure Kubernetes Service (AKS). Designed inter-service communication using REST and asynchronous messaging EDA (Azure Event Grid / Queues), ensuring distributed data consistency via Saga, CQRS, and Outbox patterns.
- **AI Orchestration into .NET Ecosystem:** Architected and deployed production enterprise RAG systems leveraging vector embeddings (`pgvector`). Built an MS Semantic Kernel plugin to integrate core software via ASP.NET 8 REST HATEOAS API, enabling secure LLM assistant integration.
- **Python-Driven LLM Infrastructure:** Conducted target R&D and executed open-source LLM fine-tuning via LoRA using the Transformers library (Python, PyTorch) to customize, optimize, and align open-source models for domain-specific enterprise workflows.
- **Interoperability:** Developed critical components in C++ to seamlessly integrate CLR-managed and native code.
- **Quality Assurance and Refactoring:** Practice DDD, SOLID, GoF. Established testing standards by implementing comprehensive unit and integration testing (xUnit, NSubstitute) and Test-Driven Development (TDD). Managed build and deployment workflows via Azure DevOps pipelines.

Samara State Technical University | Samara, Russia

Computer Vision Engineer (Part-Time / Fellowship) | September 2017 – September 2019

- **System Design:** Engineered a machine vision diagnostic system utilizing Deep Convolutional Neural Networks (CNNs) to automate the detection of broken railway fasteners and bindings using Python, PyTorch, OpenCV, TorchVision, and CUDA/cuDNN acceleration.
- **MLOps:** Gathered imaging dataset for CNN training, successfully managed severe data class imbalances by implementing synthetic generation and augmentation techniques via `imbalanced-learn` (SMOTE).
- **Integration & Async Backend:** Tied the machine vision engine directly into enterprise infrastructure via a custom REST HATEOAS API (Django REST Framework) and asynchronous messaging streams (Kafka/RabbitMQ). The successful production rollout in JSC RPC InfoTrans led to a 60% average increase in registered defect detection accuracy.

VDcom | Samara, Russia

Senior Software Specialist (Promoted for Outstanding Performance) | May 2010 – March 2014

- **Outstaffing at Sberbank CIB:** Developed backend functionality on the Diasoft platform for the investment banking business.

- **High-load Databases:** Designed, developed, and optimized complex stored procedures, ER-models and data-tier components for high-load transaction systems across Sybase ASE and MSSQL database engines.
- **Technical Estimation:** Conducted technical analysis and pre-sale engineering evaluations for resource and cost estimation.

ACS – VolgaEVMcomplex | Samara, Russia

Project Manager / .NET Developer | February 2006 – May 2010

- **Systems Analysis & Design:** Partnered with clients to formalize requirements into detailed Technical Specification Documents (TSDs) for large-scale business process automation.
- **Full-Stack C# Engineering:** Managed end-to-end Enterprise Content Management (ECM) implementations while actively writing custom C# extensions and core integration modules.

EDUCATION & RESEARCH

Samara State Technical University (SamSTU) | Samara, Russia

Ph.D. (K.T.H.) in Technical Sciences | *Specialization: Applied Neural Networks* | **Graduated 2019**

- **Thesis:** “Intelligent informational measuring system for thermography-based diagnostic of technical objects based on artificial neural networks.”
- **Research Portfolio:** Published academic insights available at ResearchGate <https://www.researchgate.net/profile/roman-girin/research> .

Samara State Technical University (SamSTU) | Samara, Russia

Engineer's Degree (5-year Specialist Degree with Honors) | **Graduated 2002**

CERTIFICATIONS & PORTFOLIO

- **Languages:** English (Full Professional Proficiency / C1), Russian (Native)
- **Certifications:** Federal Financial Markets Service (FFMS) Qualification Certificate for Financial Market Specialists (2008)
- **Portfolio:** <https://romangirin.github.io/>